Exercice 2 :

We have $\phi = \ell_1$ so

$\hat{\phi} = \hat{\ell}_1 = r_1$ the sample autocorellation

and $E[r_1] = \ell_1 = \phi$, $Var(r_1) \simeq \frac{1-\phi^2}{n}$

for $n \to \infty$,

$$\sqrt{n}\,\frac{(\ell_1 - \phi)}{\sqrt{1-\phi^2}} \sim \mathcal{N}(0,1)$$

so an $\alpha = 95\%$ confidence interval for ϕ is

$$CI = \left[\hat{\phi} - z_{\alpha/2}\sqrt{\frac{1-\phi^2}{n}},\ \hat{\phi} + z_{\alpha/2}\sqrt{\frac{1-\phi^2}{n}}\right]$$

with $z_{\alpha/2} = 1,96$

We want $(1,96)^2\left(\frac{1-\phi^2}{n}\right) \leq (0,1)^2$

this yield

$$n \geq \frac{(1,96)^2(1-0,7^2)}{0,01} = 195,9 \quad ; \quad n_{opt} = 196$$

Conclusion we need at least 196 time series observation.